

CHAPTER 1

INTRODUCTION

1.1 Introduction

In modern transportation, the suspension system is a key element for maintaining ride comfort, passenger safety, and vehicle stability. Its main role is to absorb shocks from uneven road surfaces, reduce vibrations, and ensure continuous contact between the tires and the road. Most vehicles rely on passive suspension systems, which use fixed springs and dampers. Although these systems are simple, cost-effective, and reliable, they are unable to adjust their characteristics according to varying road conditions. As a result, vehicles may suffer from reduced comfort, poor handling, and instability, especially on rough roads, sharp turns, or at high speeds.

As automotive technology continues to advance and the demand for safer, more comfortable vehicles increases, the limitations of passive suspension systems have become more evident. To address these challenges, semi-active and active suspension systems were introduced. Semi-active systems offer limited control over damping, while active systems use actuators and controllers to continuously modify suspension behavior. Despite their advantages, these systems often rely on predefined control methods such as PID or fuzzy logic, which may not be sufficiently adaptive in complex and dynamic real-world conditions.

Recent progress in Artificial Intelligence (AI) and autonomous control has opened new possibilities in vehicle systems. Intelligent systems can now process real-time sensor data, interpret environmental conditions, and make independent decisions. One emerging concept is Agentic AI, which enables systems to observe, analyze, and act without continuous human involvement. When combined with advanced techniques such as Geometric Control, suspension systems can achieve more precise and adaptive real-time responses.

Several studies have investigated active suspension control using methods like neural networks, fuzzy logic, and adaptive control. These approaches have shown noticeable improvements in ride quality and vehicle stability. However, a clear research gap exists in integrating Agentic AI-based decision-making with Geometric Control-based precision adjustment within a unified system.

Most existing solutions focus either on intelligent prediction or mechanical control, rather than combining both effectively. This highlights the need for a system that merges intelligent analysis with accurate control.

The primary issue addressed in this project is the inability of conventional suspension systems to adapt to changing road conditions, leading to reduced comfort, instability, and decreased vehicle performance. To overcome this limitation, the proposed project titled “Agentic AI Based Active Suspension and Geometric Control System” aims to develop an intelligent suspension prototype capable of automatically adjusting its behavior based on real-time road inputs.

The objectives of the project include designing an active suspension system, implementing Agentic AI for decision-making, applying Geometric Control for precise adjustments, and improving overall ride comfort and stability. The system uses sensors such as MPU6050, VL53L0X, ultrasonic sensor, and a camera module to collect real-time data. An ESP32 microcontroller acts as the central processing unit, while the BTS7960 motor driver and actuator are used to simulate suspension movement.

The technical approach involves acquiring environmental data through sensors, processing this data using the ESP32, applying AI-based logic to determine the required response, and controlling the actuator through PWM signals to adjust the suspension. This results in a closed-loop adaptive control system capable of responding effectively to changing conditions.

The scope of this report includes the design, development, and testing of the proposed intelligent suspension system. It covers hardware implementation, control strategy, system integration, and performance evaluation. Although implemented as a prototype, the concept can be extended to real vehicles, autonomous systems, and future smart transportation technologies.

1.2 Problem Statement

Conventional suspension systems are not capable of adapting to varying road conditions. They provide fixed damping and stiffness, which leads to poor ride quality on rough roads and instability at high speeds.

Therefore, there is a need to develop an intelligent system that can automatically adjust suspension parameters based on real-time conditions.

1.3. Objectives

- To design an active suspension system
- To implement Agentic AI-based control
- To improve vehicle stability and comfort
- To develop a working prototype using ESP32
- To control suspension using motor driver

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

This chapter presents an overview of recent advancements in intelligent suspension systems, focusing on techniques such as reinforcement learning, predictive suspension strategies, and sensor fusion for vehicle dynamics. The aim of this survey is to analyze existing solutions, recognize their limitations, and emphasize the necessity for an integrated approach like the **Agentic AI Based Active Suspension and Geometric Control System**.

2.2 Reference 1

A. Baba Ahmadi, M. Sadeghi, and H. Lee, “A Deep Reinforcement Learning-Based Controller for Magnetorheological-Damped Vehicle Suspension,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 5, pp. 6784–6795, 2023.

This research introduces a suspension control method based on Deep Reinforcement Learning (DRL) for vehicles using magnetorheological dampers. The controller learns optimal control strategies by interacting with changing road conditions and adapting its response accordingly. Through continuous adjustment of damping parameters, the system achieves improved ride comfort, reduced vibration, and enhanced vehicle handling compared to conventional methods. The study highlights the effectiveness of learning-based techniques in improving the adaptability of suspension systems under dynamic driving environments.

2.3 Reference 2

Y. Li, J. Zhang, and X. Huang, “Adaptive AI-Based Semi-Active Suspension System for Terrain Classification and Ride Comfort Enhancement,” *Journal of Mechanical Systems and Signal Processing*, vol. 188, pp. 109–123, 2022.

This work presents an AI-based semi-active suspension system that incorporates terrain classification to distinguish between various road conditions such as smooth surfaces, rough terrain, and speed breakers. Based on the identified terrain type, the system adjusts its damping characteristics to improve comfort and stability. Experimental results demonstrate that the proposed system provides better adaptability and performance when compared to traditional semi-active suspension systems.

2.4 Conclusion from References

From the reviewed literature, it can be concluded that suspension systems utilizing Artificial Intelligence and Reinforcement Learning offer enhanced ride comfort, improved stability, and better adaptability than conventional passive systems. However, most of the existing methods concentrate either on damping control or terrain classification individually. There is still a gap in combining Agentic AI-based autonomous decision-making with Geometric Control-based precision adjustment within a single system. Integrating these approaches can result in a more efficient and real-time adaptive suspension system suitable for future smart vehicle applications.

CHAPTER 3

PROJECT SYSTEM DESIGN

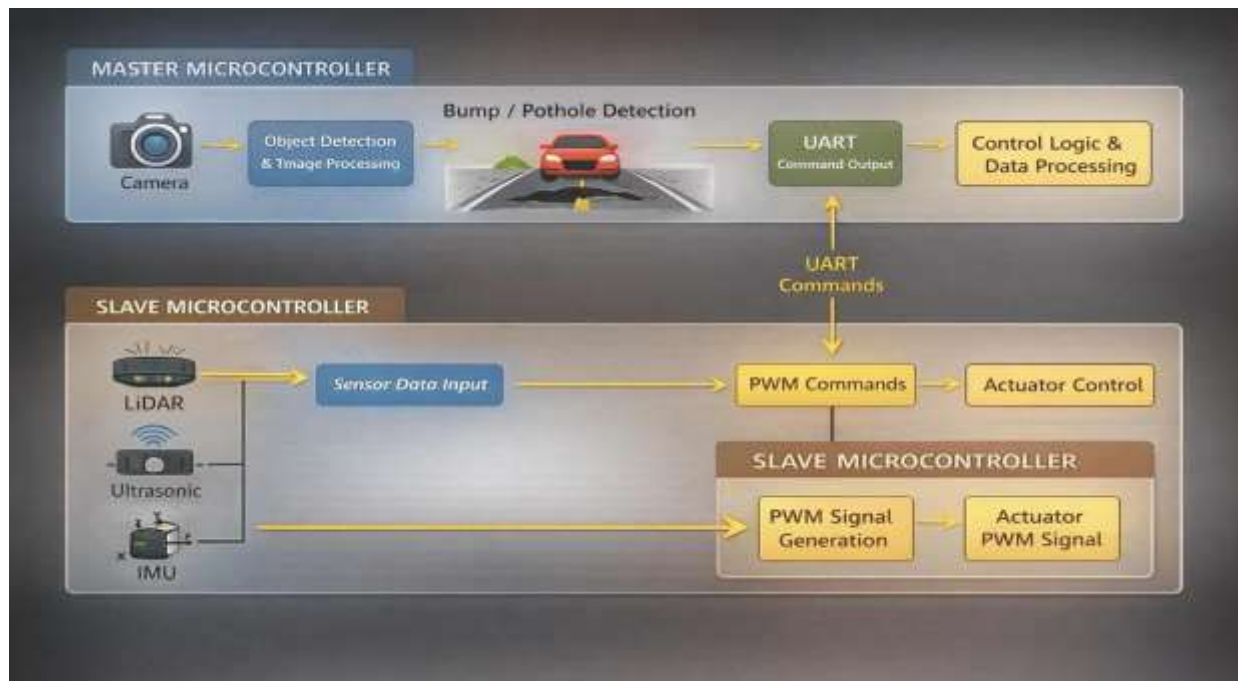
Introduction

This chapter outlines the overall design and architecture of the proposed system, including the block diagram, system operation, component selection, and specifications. It explains how both hardware and software modules are combined to achieve intelligent suspension control using Agentic AI and Geometric Control methods.

3.1 Block Diagram of System and Working of the System

The main objective of this project is to develop an intelligent suspension system that can automatically respond to changing road conditions, thereby enhancing ride comfort and vehicle stability. The system integrates multiple sensors, ESP32 controllers, AI-based decision-making, and actuator control to enable real-time suspension adjustment.

Fig. 3.1 Block Diagram of Proposed System



Working of the System

1. Power Supply Block

The system operates using a battery or external power source. Voltage regulators such as LM317 and 7805 are used to provide a steady and regulated supply to all components, including the ESP32, sensors, and motor driver. This ensures stable and safe operation of the system.

2. OV3640 Camera

The OV3640 camera captures real-time images of the road ahead. These images help in detecting irregularities such as bumps or obstacles. The captured data is then forwarded to the Master ESP32 for processing.

3. Master ESP32

The Master ESP32 acts as the initial processing unit. It receives input from the camera, performs preprocessing, and sends the processed data to the Slave ESP32 for further analysis.

4. Slave ESP32

The Slave ESP32 functions as the main decision-making controller. It collects data from multiple sources, including:

- Master ESP32
- MPU6050 vibration sensor
- VL53xx distance sensor
- Ultrasonic sensor

Based on these inputs, it applies Agentic AI logic to determine the appropriate suspension response.

5. Sensors Block

The system uses multiple sensors to gather real-time environmental data:

- MPU6050 (Vibration Sensor): Detects vibrations, shocks, and tilt.
- VL53xx Sensor: Measures distance or displacement for suspension positioning.
- Ultrasonic Sensor: Identifies obstacles and uneven surfaces.

All sensor data is continuously transmitted to the Slave ESP32 for processing and control decisions.

6. Motor Driver (BTS7960)

The BTS7960 motor driver receives control signals from the Slave ESP32 and regulates the motor's speed and direction. It is capable of handling high current, making it suitable for actuator applications.

7. Linear Actuator

The linear actuator converts electrical signals into mechanical motion, enabling physical adjustment of the suspension system.

8. Suspension Motor

The suspension motor acts as the final output stage. It adjusts the suspension characteristics by making it softer or stiffer based on the control signals generated by the system.

3.2 Hardware Description

The hardware setup of the proposed system includes a microcontroller, motor driver, power supply unit, and supporting electronic components. Each part contributes to the overall performance and reliability of the system.

ESP32 Microcontroller



Fig 3.2.1 ESP32

The ESP32 is the central control unit responsible for processing sensor data and executing Agentic AI logic. It generates control signals for the motor driver and is selected due to its high processing speed, low power consumption, and integrated communication features such as Wi-Fi and Bluetooth.

- Operating Voltage: 3.3V (logic level)
- Input Voltage: 5V (via USB) or 3.3V (direct supply)
- Processor: Dual-core Tensilica Xtensa LX6
- Clock Speed: Up to 240 MHz
- Wi-Fi: 802.11 b/g/n (built-in)
- Bluetooth: Bluetooth v4.2 (Classic + BLE)
- GPIO Pins: ~30–36 programmable pins
- Analog Input Pins: 12-bit ADC

BTS7960 Motor Driver



Fig 3.2.2 B TS7960 Motor Driver

The BTS7960 is a high-current H-bridge motor driver used to control the actuator. It receives control signals from the ESP32 and manages motor speed and direction. Its high current handling capability makes it suitable for suspension applications.

- Driver Type: H-Bridge Motor Driver (IBT-2 Module)
- Operating Voltage (Motor): 5V to 27V
- Logic Voltage: 3.3V – 5V compatible
- Maximum Current: Up to 43A
- Control Method: PWM (Pulse Width Modulation)
- Control Pins: RPWM, LPWM, R_EN, L_EN
- Switching Frequency: Up to 25 kHz

LM317 Voltage Regulator



Fig 3.2.3LM317 Voltage Regulator

- Type: Adjustable Voltage Regulator
- Output Voltage Range: 1.25V to 37V
- Input Voltage: Up to 40V (max)
- Output Current: Up to 1.5A
- Reference Voltage: 1.25V
- Adjustment Method: External resistors (R1, R2)
- Ripple Rejection: Good (~80 dB)
- Package Type: TO-220
- Temperature Range: 0°C to 125°C

7805 Voltage Regulator



Fig 3.2.4 7805 Voltage Regulator

- Type: Fixed Voltage Regulator
- Output Voltage: 5V (constant)
- Input Voltage: 7V to 35V
- Output Current: Up to 1A (with proper heatsink)
- Dropout Voltage: ~2V
- Regulation: Line and load regulation available
- Package Type: TO-220
- Operating Temperature: 0°C to 125°C

Voltage regulation is achieved using LM317 and 7805 to provide stable power. The LM317 offers adjustable output, while the 7805 supplies a fixed 5V output. These regulators ensure safe operation of sensitive components like ESP32 and sensors.

Batteries (DMEGC 2600mAh)



Fig 3.2. 5Batteries (DMEGC 2600mAh)

The system uses DMEGC 2600mAh lithium-ion batteries as the main power source. Multiple batteries are used to meet the voltage and current requirements of the system, ensuring continuous and portable operation.

- Type: Lithium-ion Rechargeable Battery (18650)
- Capacity: 2600mAh
- Nominal Voltage: 3.7V per cell
- Fully Charged Voltage: 4.2V
- Cut-off Voltage: ~3.0V
- Rechargeable: Yes
- Cycle Life: ~300–500 charge cycles
- Energy Density: High (compact and lightweight)
- Application: Provides power supply to the complete system

Supporting Components

Additional components such as resistors, capacitors, wires, zero PCB, solder wire, and flux are used for circuit assembly and stability. Resistors help regulate current, while capacitors reduce noise and stabilize voltage. The zero PCB provides a permanent and organized circuit structure.

2.1 Selection of Components

Table 3.1 Comparative Analysis of Components

Sr. No.	Component	Available Options	Selected Component	Justification
1	Microcontroller	Arduino Uno, Raspberry Pi, ESP32	ESP32	High speed, Wi-Fi, Bluetooth, multiple GPIO pins
2	Motor Driver	L298N, BTS7960, Relay Module	BTS7960	High current handling (43A), PWM support
3	Motion Sensor	ADXL345, MPU6050	MPU6050	Accelerometer + Gyroscope in one module
4	Distance Sensor	IR Sensor, VL53L0X	VL53L0X	High accuracy, laser-based ToF sensing
5	Voltage Regulator	7805, LM317, Buck Converter	LM317 + 7805	Adjustable + fixed regulated output
6	Power Source	Lead Acid, Li-ion, Adapter	Li-ion Battery	Rechargeable, compact, lightweight

Justification for Selection

- The **ESP32** is selected for its strong processing performance, energy efficiency, and integrated wireless features such as Wi-Fi and Bluetooth, which make it suitable for real-time control applications.
- The **BTS7960 motor driver** is chosen due to its ability to handle high current loads while providing smooth and reliable control over motor speed and direction.
- The **MPU6050** is preferred as it combines accelerometer and gyroscope functionalities in a single module, allowing accurate measurement of vibration, motion, and tilt.
- The **VL53L0X sensor** is selected for its high precision in distance measurement using Time-of-Flight (ToF) technology.

- The **LM317 and 7805 voltage regulators** are used to maintain a stable power supply, offering both adjustable and fixed voltage outputs for different system components.
- The **Li-ion battery** is chosen because of its compact design, rechargeable nature, and high energy density, making it ideal for portable and continuous operation.

2.2 Specifications of Project

The important specifications of the proposed system are as follows:

- **Microcontroller:** ESP32 dual-core processor operating up to 240 MHz
- **Operating Voltage:** 3.3V logic level
- **Motor Driver Capacity:** Supports up to 43A current
- **Output Voltage:** Regulated 5V along with adjustable voltage supply
- **Battery Capacity:** 2600mAh Lithium-ion battery
- **Communication Protocols:** I2C, PWM, and digital input/output
- **Sensors Used:** MPU6050, VL53L0X, and camera module
- **Control Technique:** Combination of Agentic AI and Geometric Control
- **Application:** Active suspension control and adjustment
- **Operation Mode:** Real-time adaptive control based on environmental conditions

CHAPTER 4

DEVELOPMENT AND TESTING

4.1 Testing

Testing was carried out to verify the performance of different components and the overall working of the project after hardware implementation. The testing process helped in analyzing sensor response, motor operation, control accuracy, and system behavior under different road conditions.

4.2 Testing of Individual Components

1. Power Supply Testing

- LM317 and 7805 regulators were tested using a multimeter.
- Stable output voltages were obtained for controller and sensors.

2. Sensor Testing

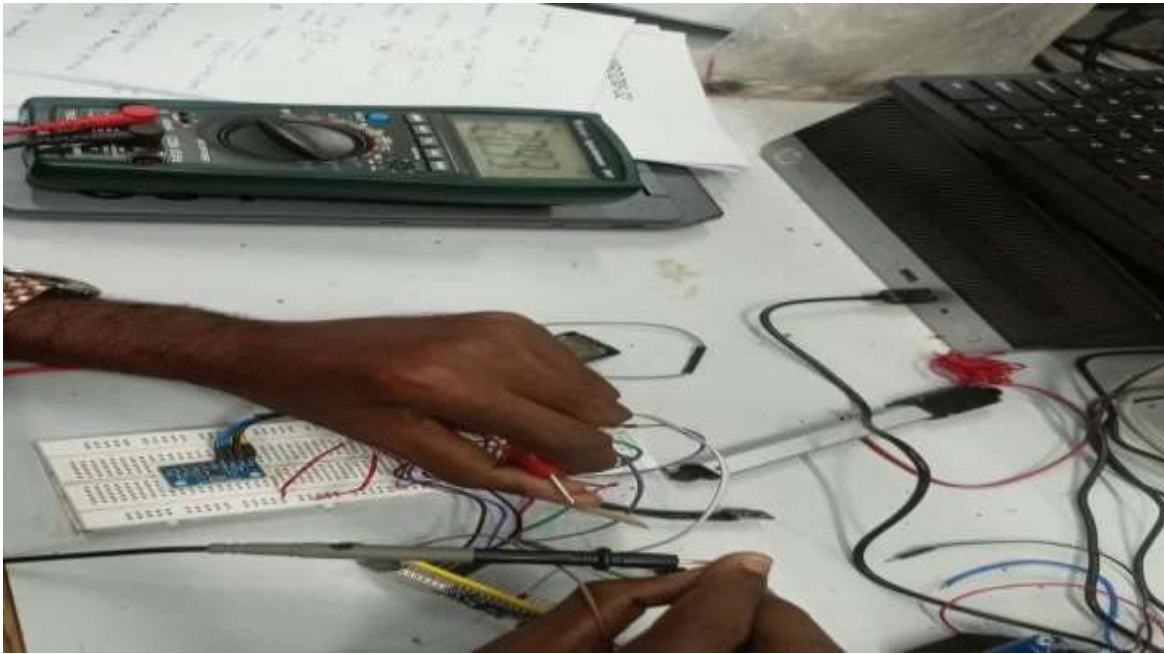


Fig 4.2.1 MPU6050 Sensor Testing

- MPU6050 was tested for vibration and tilt detection.

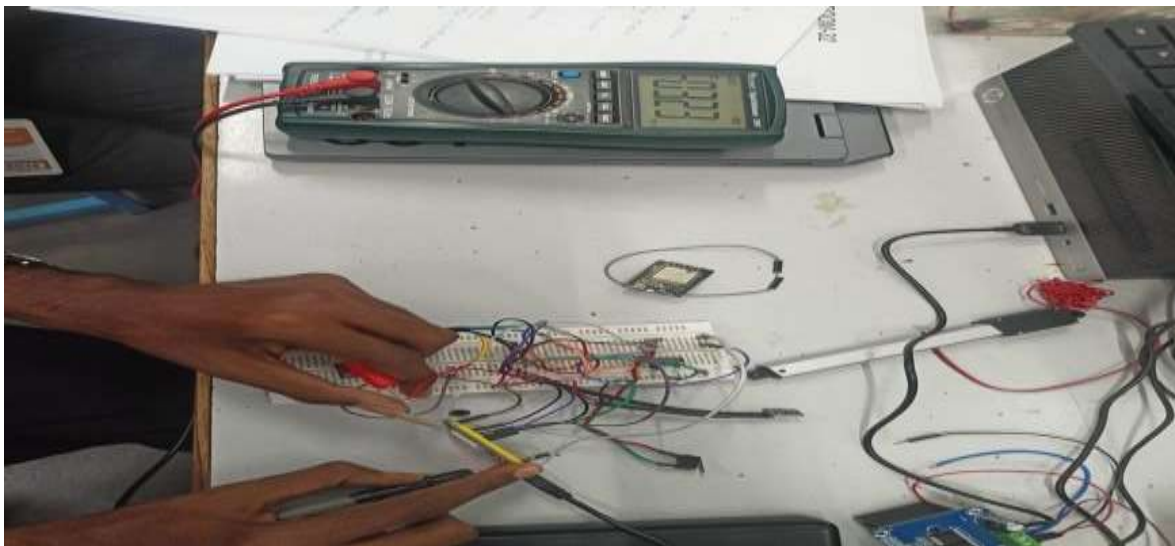


Fig 4.2.2 VL53L0X Sensor Testing

- VL53L0X and ultrasonic sensors were tested for distance measurement accuracy.
- Camera module was tested for capturing road condition image

3. Motor Driver Testing

- BTS7960 driver was tested using PWM signals from ESP32.
- Forward, reverse, and speed control of motor were verified successfully.

4. Controller Testing

- BTS7960 driver was tested using PWM signals from ESP32.
- Forward, reverse, and speed control of motor were verified successfully.

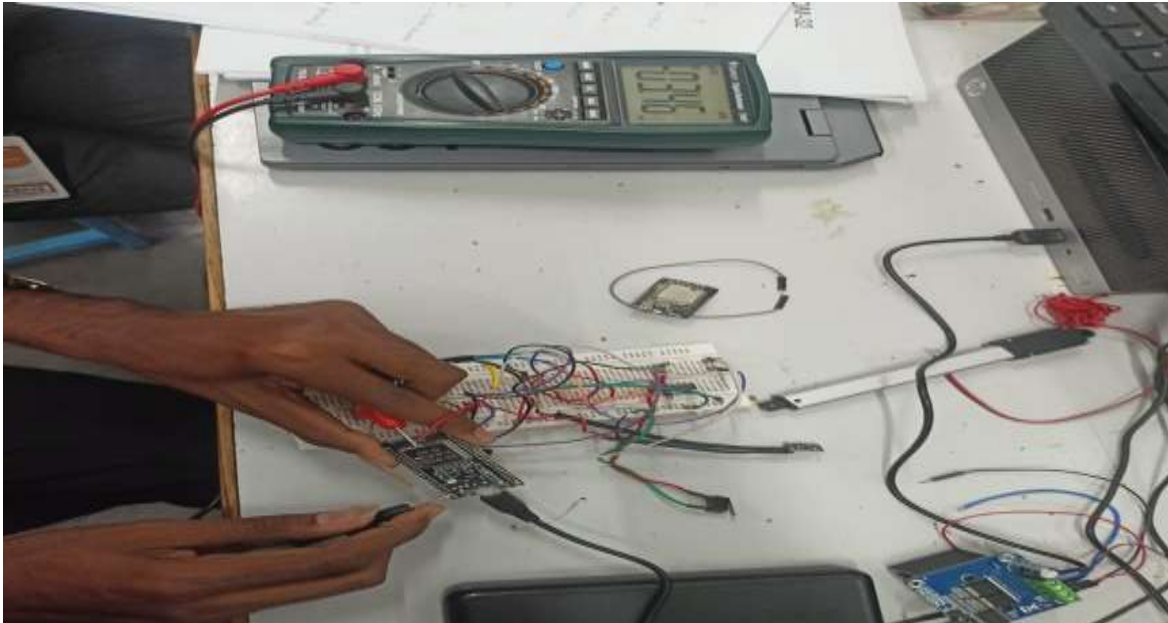


Fig 4.2.3 ESP 32

- ESP32 program execution and sensor data processing were tested.
- Correct decision-making and control signal generation were observed.

Complete System Testing

The complete system was tested by integrating sensors, controller, motor driver, and actuator. The prototype successfully adjusted the suspension position according to detected vibration and road conditions. Under rough conditions, the actuator responded quickly to reduce vibration, while under smooth conditions stable operation was maintained.

Thus, the testing results confirmed the proper functioning and reliability of the proposed intelligent

Power Consumption Analysis of Project Components

The table below presents the voltage, current, and power consumption values for each component utilized in the Agentic AI Based Active Suspension and Geometric Control System. The overall power requirement of the system is estimated to be around 20 Watts.

Table 4.1 Power Consumption of Components

Sr. No.	Component Name	Operating Voltage (V)	Current (A)	Power (W)	Description
1.	ESP32 Dev Board	5V (3.3V logic)	0.35 A	1.75 W	Main controller for processing sensor data and control signals
2.	MPU6050 Sensor	5V	0.07 A	0.35 W	Detects vibration, tilt, and motion
3.	LiDAR Sensor (VL53L0X)	3.3V	0.036 A	0.12 W	Measures distance using Time-of-Flight technology
4.	BTS7960 Motor Driver + Actuator	12V	1.2 A	14.40 W	Controls actuator movement for suspension adjustment
5.	ESP32-CAM	5V	0.295 A	1.475 W	Captures road images for condition analysis

Total Power Consumption

Parameter	Value
Total Power	18.09W
Recommended	20W Minimum

Detailed Analysis

1. ESP32 Dev Board

- Operates with a 5V input supply while internally using 3.3V logic levels.
- Draws approximately 350 mA during data processing and communication tasks.
- Functions as the primary controller of the entire system.

2. MPU6050 Sensor

- A low-power module designed for motion sensing applications.
- Capable of detecting vibration, acceleration, and tilt.
- Consumes minimal power, approximately 0.35 W.

3. VL53L0X LiDAR Sensor

- A laser-based distance measurement sensor with very low current requirements.
- Provides high accuracy while maintaining low power consumption.

4. BTS7960 Motor Driver + Actuator

- Represents the highest power-consuming unit in the system.
- Responsible for driving the actuator motor used in suspension adjustment.
- Requires a stable 12V power supply for proper operation.

5. ESP32-CAM

- Utilized for capturing real-time road conditions ahead of the vehicle.
- Consumes relatively higher current due to camera functionality and Wi-Fi transmission.

Troubleshooting Steps

The following troubleshooting steps should be followed if the system does not work properly:

Table 4.2 Troubleshooting Guide

Step No.	Problem Check	Action
1	Check all circuit connections	Use continuity tester or multimeter to verify wiring connections
2	Check power supply	Verify voltage levels (Vcc, GND) and ensure common ground connection
3	Check I2C communication	Verify SDA and SCL lines for MPU6050 and VL53L0X sensors
4	Check limit switch / actuator movement	Ensure actuator moves freely and switches are functioning properly
5	Check UART cable	Verify TX/RX serial cable connection for communication
6	Check Serial Monitor	Open Arduino IDE Serial Monitor and verify baud rate
7	Restart system	Reset ESP32 and power cycle the complete system
8	Re-upload code	Upload the latest firmware/code again if required

Conclusion

The power analysis indicates that the motor driver and actuator account for the highest power consumption in the system, whereas the sensors operate with minimal power requirements. A regulated power supply of approximately 20 W is adequate to ensure stable system operation. Additionally, implementing proper troubleshooting procedures contributes to maintaining consistent and reliable performance of the system.

4.3 PCB Artwork

The PCB artwork for the project was designed to provide compact and reliable circuit implementation. The PCB layout was prepared using Proteus / KiCad / EasyEDA software tool.

The design includes proper placement of:

- ESP32 controller section
- Voltage regulator circuit
- Sensor connector pins
- Motor driver connections
- Power supply tracks
- Ground plane for noise reduction

The PCB artwork ensures neat wiring, reduced signal interference, and easier hardware assembly.

4.3 Layout Versus Schematic Verification Report

After completion of the PCB layout, verification was carried out to compare the PCB design with the original schematic circuit. This process ensures that all electrical connections in the schematic are correctly represented in the PCB layout.

Verification Results:

- All component pins matched correctly with schematic connections.
- Power supply and ground tracks were properly routed.
- No open circuit or short circuit errors were found.
- Signal paths for sensors and motor driver were verified successfully.
- PCB layout was consistent with the designed schematic.

CHAPTER 5

RESULTS

5.1 Introduction

This chapter summarizes the results obtained after the development and testing of the proposed Agentic AI Based Active Suspension and Geometric Control System. It presents the system performance in the form of tabulated results along with photographs of the developed hardware prototype.

5.2 Results

The overall system was evaluated under various operating conditions, including vibration sensing, detection of uneven surfaces, obstacle identification, and actuator performance. The system was able to process sensor inputs effectively and adjust the suspension automatically in real time based on the detected conditions.

Table 5.1 Results of Complete System

Sr. No.	Test Condition	Input Detected	System Response	Status
1	Power ON Test	Supply Voltage Applied	System initialized successfully	Pass
2	MPU6050 Test	Vibration / Tilt Detected	Data sent to ESP32 correctly	Pass
3	VL53L0X Test	Distance Change Detected	Accurate distance measured	Pass
4	Ultrasonic Test	Obstacle Detected	Obstacle distance displayed / processed	Pass

Sr. No.	Test Condition	Input Detected	System Response	Status
5	Camera Test	Road Image Captured	Image data processed successfully	Pass
6	Motor Driver Test	PWM Signal Applied	Motor speed & direction controlled	Pass
7	Actuator Test	Control Signal Generated	Suspension position adjusted	Pass
8	Rough Road Condition	High vibration detected	Soft suspension response activated	Pass
9	Smooth Road Condition	Normal surface detected	Stable suspension maintained	Pass
10	Full System Test	All modules active	Real-time adaptive control achieved	Pass

5.3 Photographs

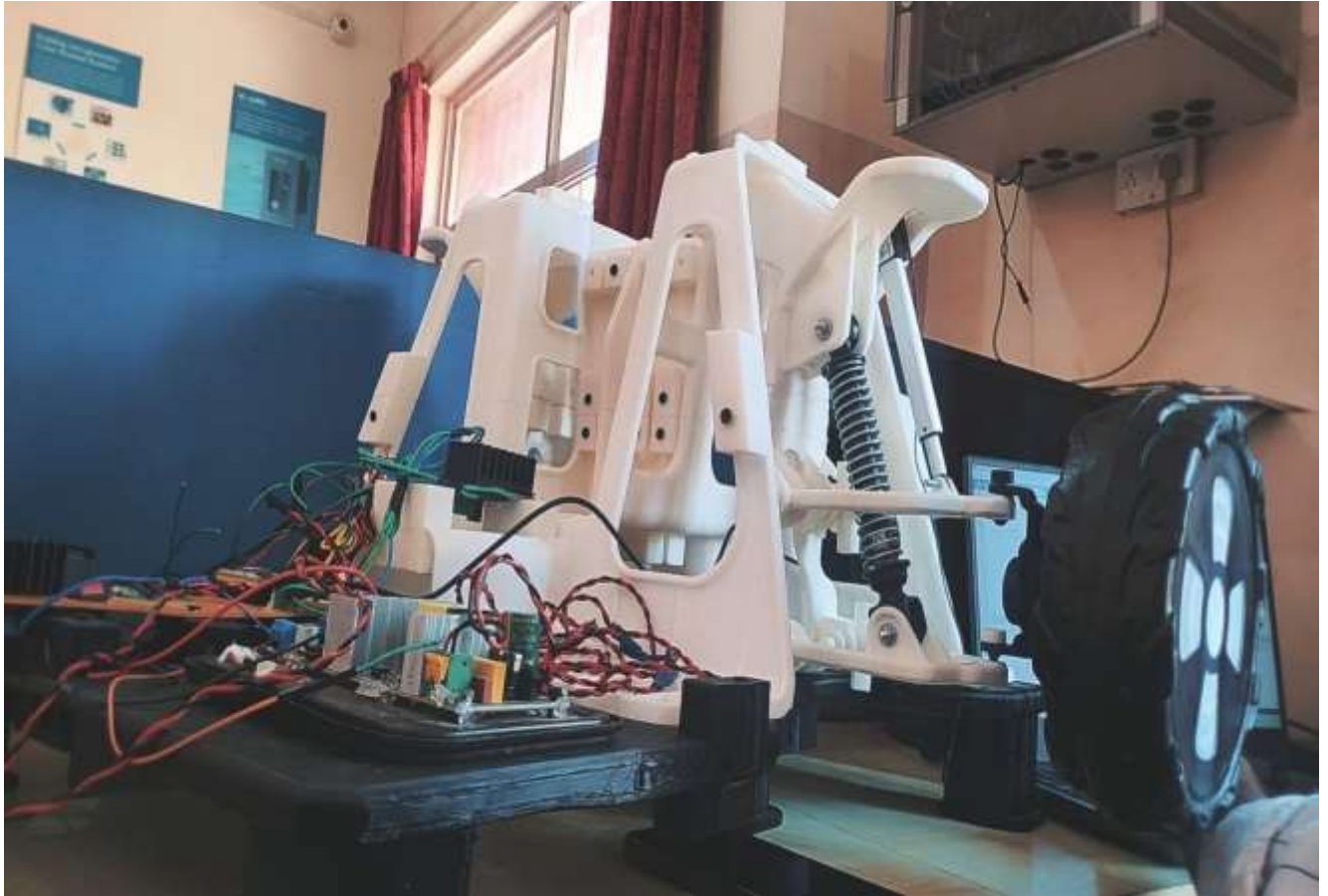


Fig. 5.1 Complete Hardware Prototyp

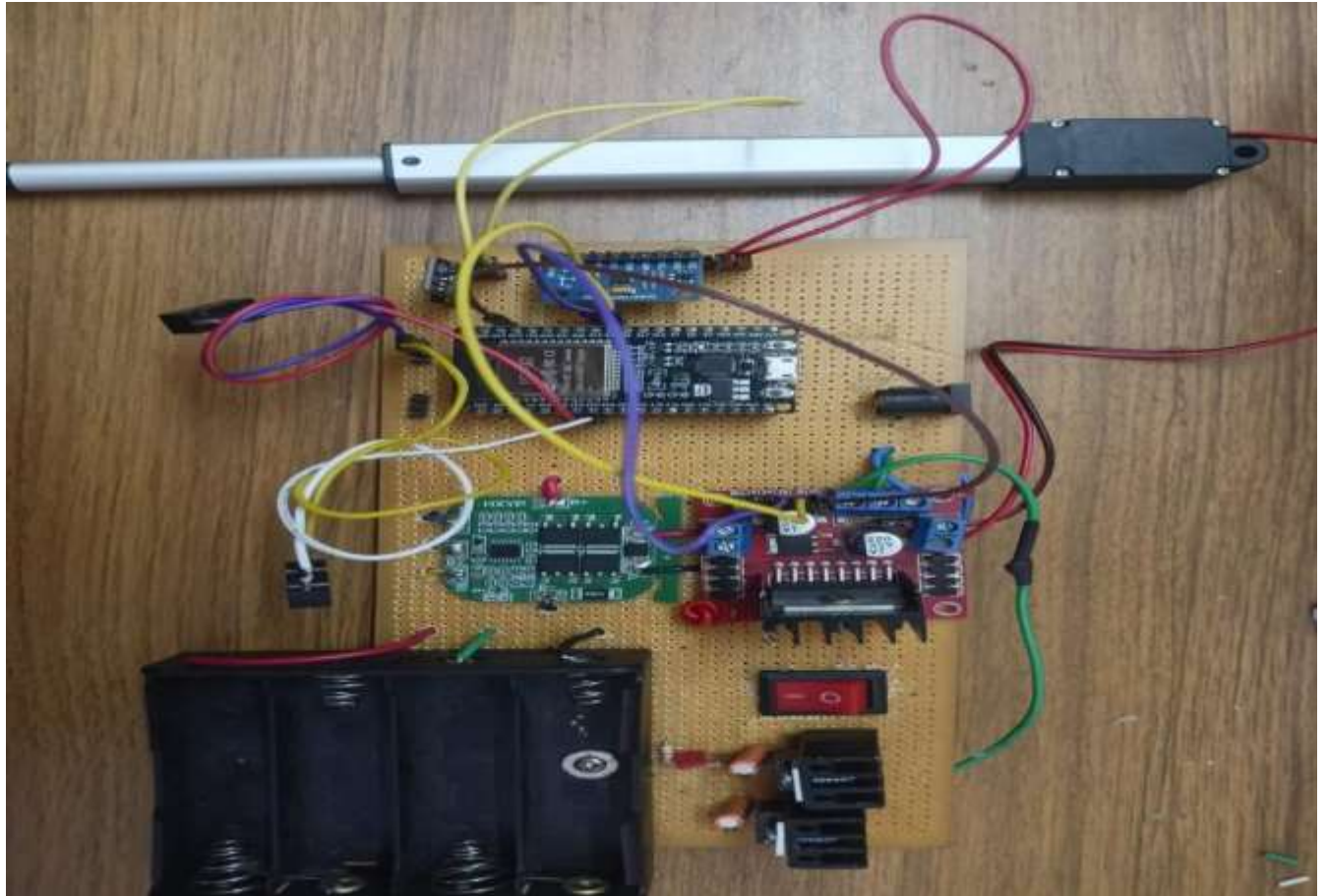


Fig. 5.2 PCB / Zero PCB Implementation

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion :

The project titled “Agentic AI Based Active Suspension and Geometric Control System” has been successfully designed and implemented to enhance vehicle ride comfort, stability, and adaptability under different road conditions. By integrating intelligent decision-making with modern control techniques, the proposed system offers improved performance compared to conventional passive suspension systems.

In this work, an ESP32-based control unit was utilized to process real-time data obtained from multiple sensors, including MPU6050, VL53L0X, ultrasonic sensor, and camera module. Based on these inputs, Agentic AI logic was applied to analyze road conditions and make autonomous decisions. Additionally, Geometric Control was incorporated to achieve precise and smooth adjustment of the suspension mechanism. The motor driver and actuator effectively simulated suspension movement in accordance with the generated control signals.

Major Achievements:

- Developed a functional prototype of an active suspension control system.
- Enabled real-time processing of sensor data using the ESP32 controller.
- Implemented intelligent decision-making using Agentic AI concepts.
- Enhanced suspension performance under vibration and uneven road conditions.
- Achieved accurate actuator control using geometric control techniques.
- Designed a scalable system suitable for future automotive applications.

Through this project, key concepts such as embedded systems, sensor interfacing, motor control, PWM techniques, artificial intelligence, and control systems were learned and applied practically. It also strengthened understanding in hardware integration, software development, and system design.

The proposed system has potential applications in smart vehicles, autonomous cars, off-road vehicles, military transportation, and advanced driver assistance systems (ADAS). Furthermore, it can be extended for real-world implementation by incorporating IoT connectivity and machine learning algorithms.

6.2 Future Scope :

The proposed Agentic AI Based Active Suspension and Geometric Control System provides a strong foundation for future intelligent vehicle technologies. Although the current project is developed as a prototype model, it has wide scope for further improvement and real-world implementation.

The future enhancements of this project are as follows:

- Integration of advanced sensors such as LiDAR, GPS, and high-resolution cameras for more accurate road condition detection.
- Implementation of Machine Learning and Deep Learning algorithms for predictive suspension control based on previous road patterns and driving behavior.
- Development of a real vehicle prototype for testing under practical driving conditions.
- Addition of IoT connectivity for remote monitoring, diagnostics, and real-time performance analysis.
- Use of cloud computing for storing sensor data and optimizing suspension performance.
- Improvement in energy efficiency by using smart power management techniques.
- Application in autonomous and electric vehicles for enhanced passenger comfort and safety.

CHAPTER 7 BILL OF MATERIAL

Table 7.1 Bill of Material

Sr No	Component Name	Use		Price
1	VL53L0x	Distance Sensor	URL	<u>114</u>
2	MPU6050	Vibration Sensor	URL	<u>293</u>
3	ESP32-WROOM-32	Microcontroller	URL	<u>363</u>
4	100mm, 15mm/s, 64N, 12V, Linear Actuator	Suspension Movement	later	<u>2000-3500</u>
5	L298N DC Motor Driver	To Drive Motor	URL	<u>188</u>
6	ESP-32 Cam (OV2640)	Camera	URL	<u>622</u>
7	60rpm 12v low noise DC Motor	Toe Adjustment	later	300-400
8	DMEGC 18650-26E 3.7V 2600mAh Li-Ion Battery Cell	Power Supply	Offline	440-500
9	18650 Lithium Polymer Battery Holder	Battery Holder	URL	<u>40</u>
10	4S - BMS	Battery Assembly	URL	<u>154</u>
11	TP - 4056 (4)	Battery Charging	URL	<u>56</u>
12	Single Cell Holder (2)	Battery Assembly	URL	<u>64</u>
13	Buck Converter (12V, 2A) and (15v, 1A)	Battery Assembly	Offline	300-400
			Total	<u>6194</u>

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